

PROJECT 4 DHCP AND RELAY

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TASK 1: CREATING STATIC ROUTES IN THE ROUTER

ROUTER 1

R1 IP: 172.50.0.1 – eth0/0

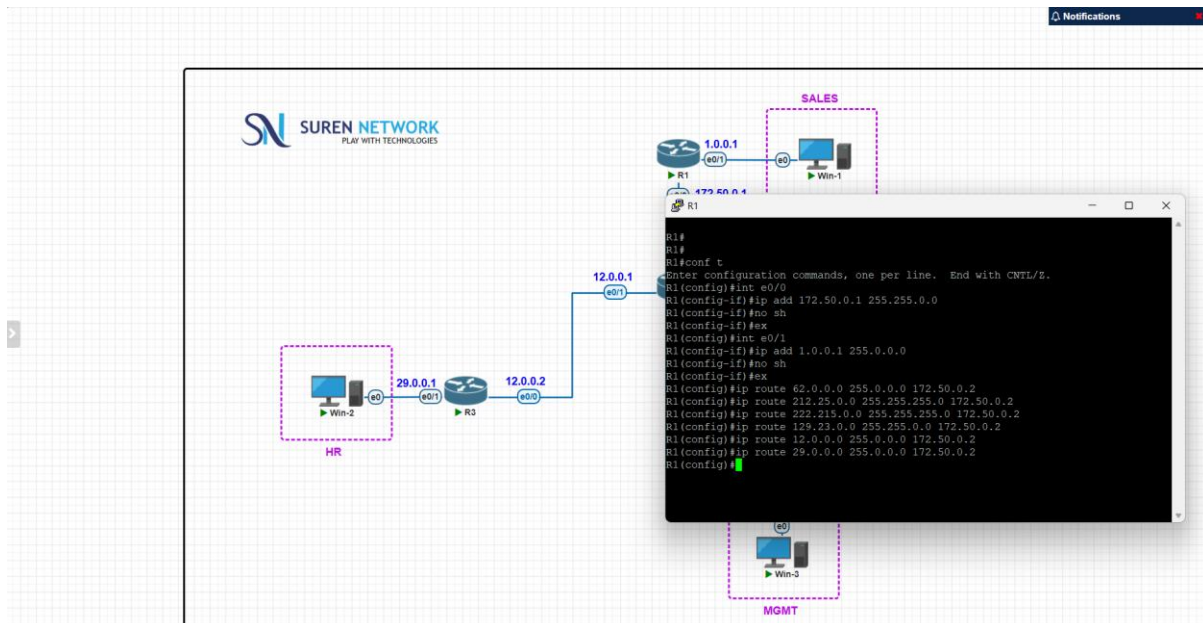
STATIC ROUTE R1:

12.0.0.0/8	172.50.0.2
29.0.0.0/8	172.50.0.2
62.0.0.0/8	172.50.0.2

R1 IP: 1.0.0.1 – eth0/1

STATIC ROUTE R1:

129.23.0.0/16	172.50.0.2
212.25.0.0/24	172.50.0.2
222.215.0.0/24	172.50.0.2



ROUTER 2

R2 IP: 172.50.0.2 – eth0/0

R2 IP: 222.215.0.0.1 – eth0/2

STATIC ROUTE R2:

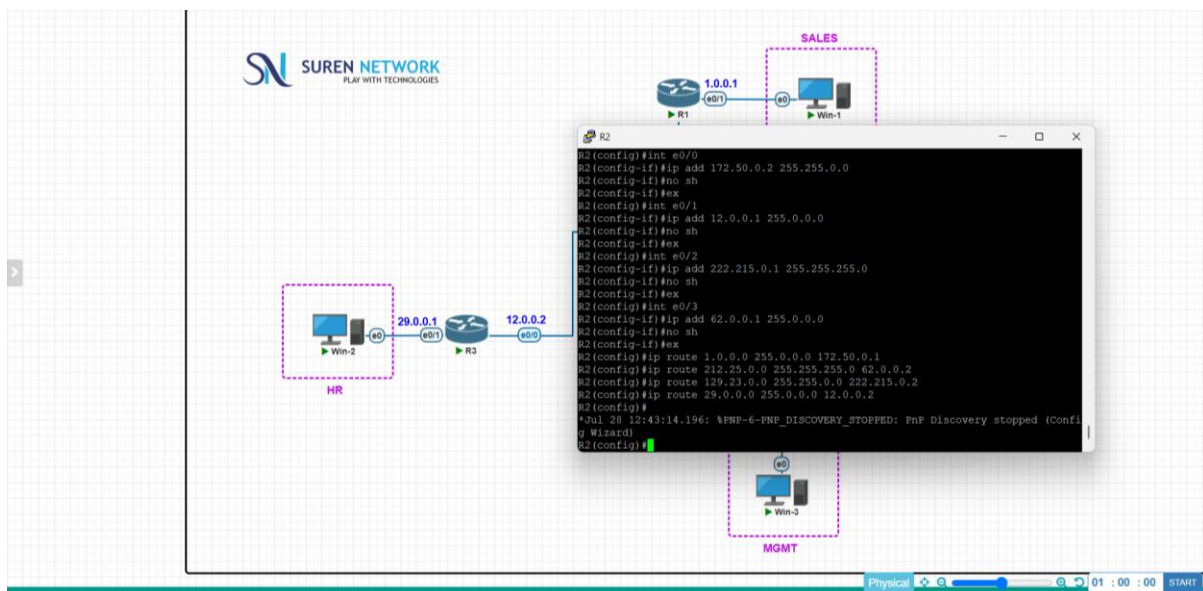
1.0.0.0/8	172.50.0.1
29.0.0.0/8	12.0.0.1

R2 IP: 12.0.0.1 – eth0/1

R2 IP: 62.0.0.1 – eth0/3

STATIC ROUTE R2:

129.23.0.0/16	225.215.0.2
212.25.0.0/24	62.50.0.2



ROUTER 3

R3 IP: 12.0.0.2 – eth0/0

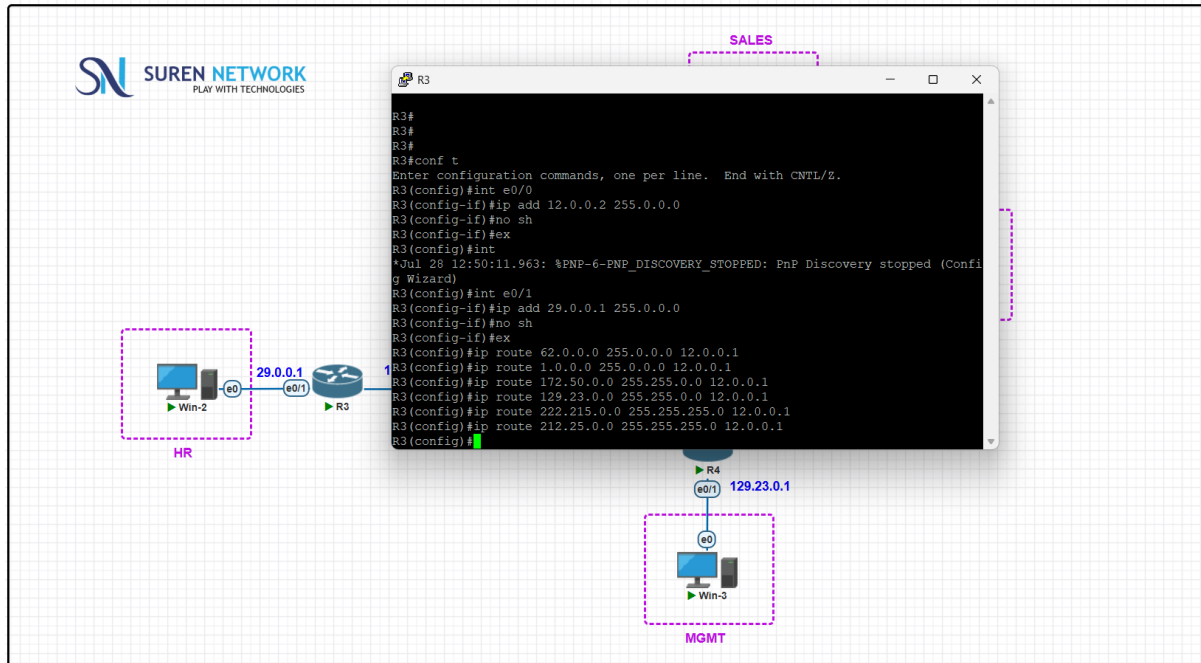
STATIC ROUTE R3:

1.0.0.0/8	12.0.0.1
62.0.0.0/8	12.0.0.1
129.23.0.0/16	12.0.0.1

R3 IP: 29.0.0.1 – eth0/1

STATIC ROUTE R3:

172.50.0.0/16	12.0.0.1
212.25.0.0/24	12.0.0.1
222.215.0.0/24	12.0.0.1



ROUTER 4

R4 IP: 222.215.0.2 – eth0/0

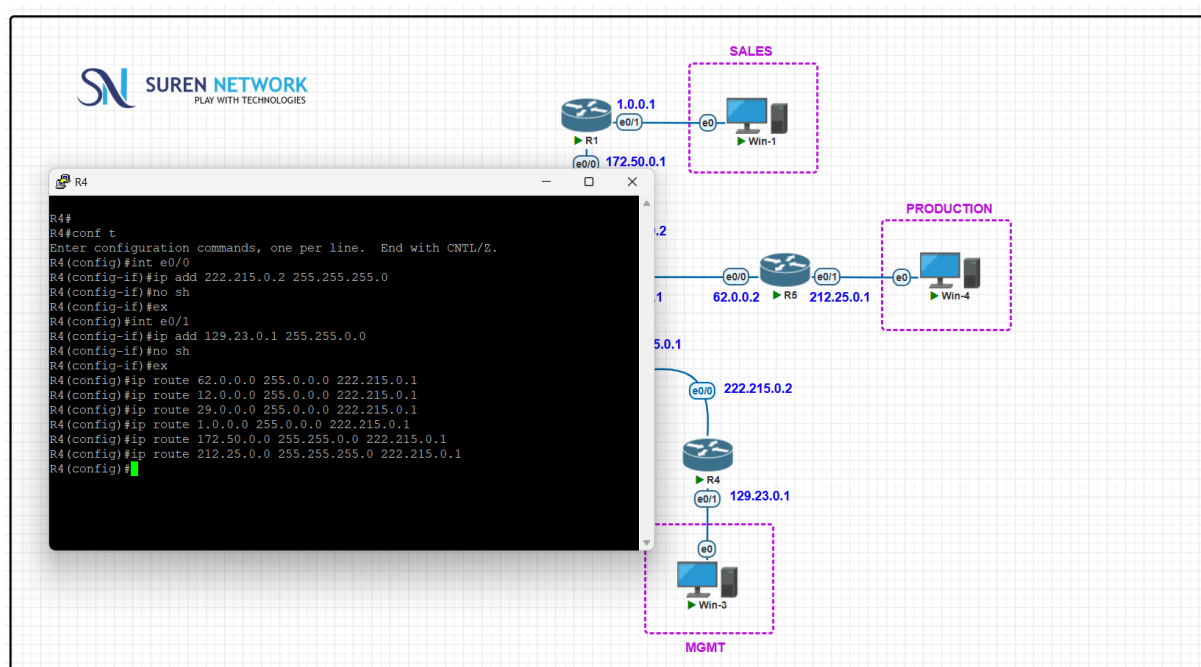
STATIC ROUTE R4:

1.0.0.0/8	222.215.0.1
12.0.0.0/8	222.215.0.1
29.0.0.0/8	222.215.0.1

R4 IP: 129.23.0.1 – eth0/1

STATIC ROUTE R4:

62.0.0.0/8	222.215.0.1
172.50.0.0/16	222.215.0.1
212.25.0.0/24	222.215.0.1



ROUTER 5

R5 IP: 62.0.0.2 – eth0/0

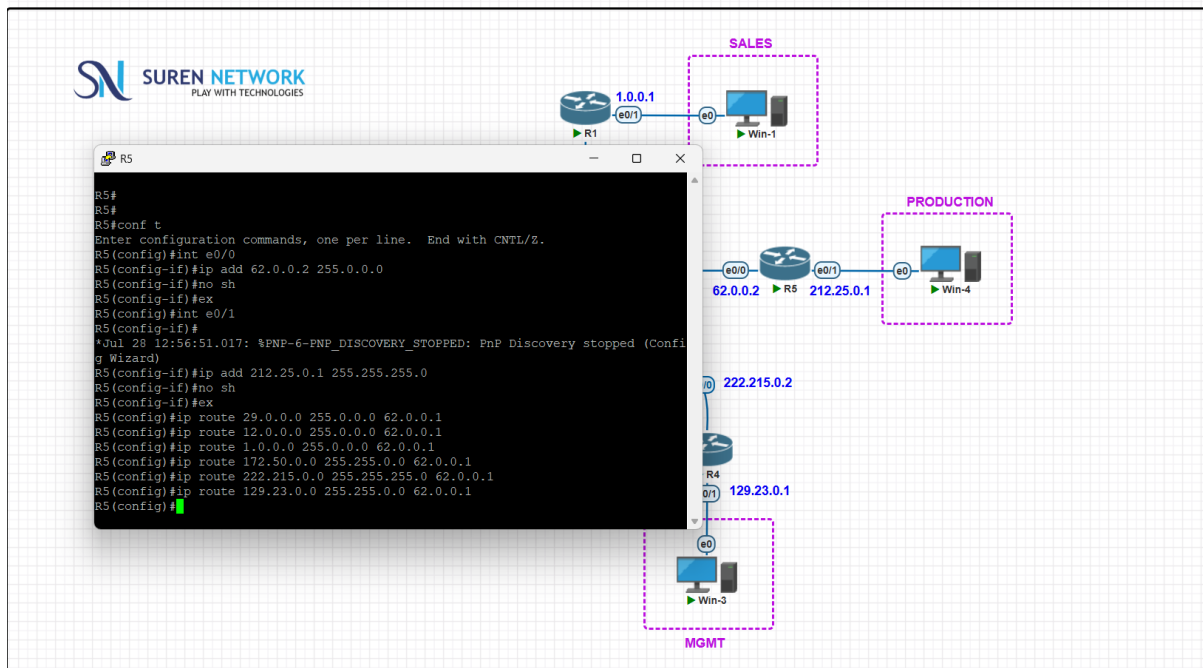
STATIC ROUTE R5:

1.0.0.0/8 62.0.0.1
12.0.0.0/8 62.0.0.1
29.0.0.0/8 62.0.0.1

R5 IP: 212.25.0.1 – eth0/2

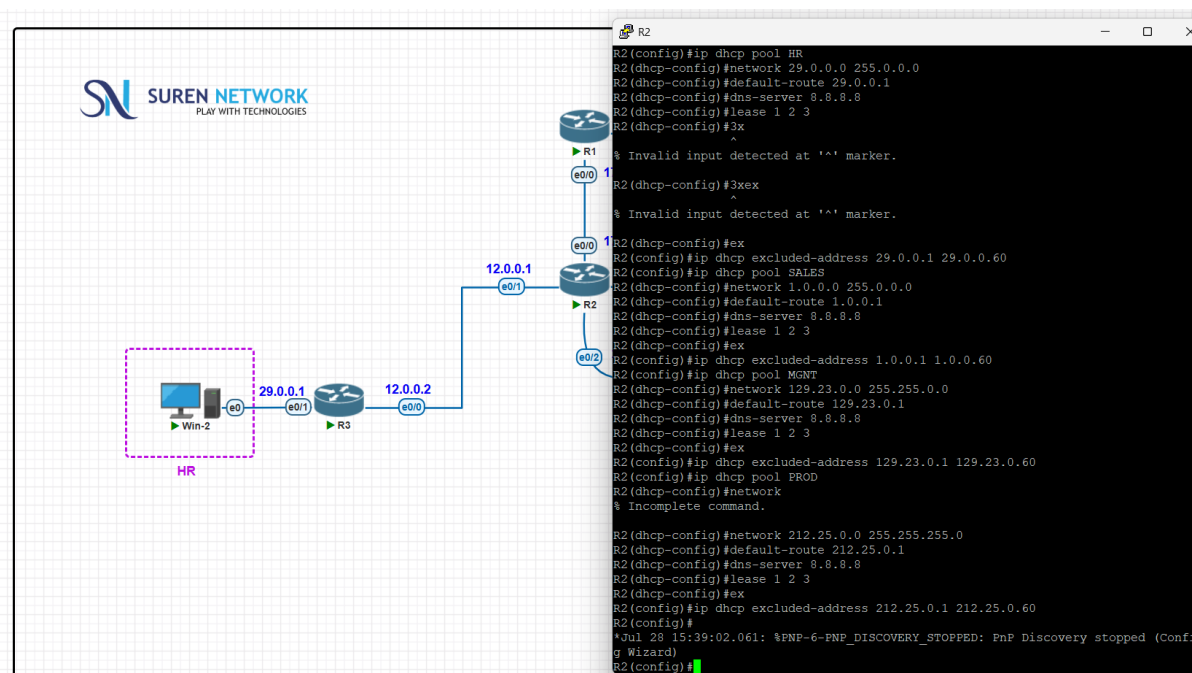
STATIC ROUTE R5:

129.23.0.0/16 62.0.0.1
172.50.0.0/16 62.0.0.1
222.215.0.0/24 62.0.0.1



TASK 2: CONFIGURING DHCP SERVICE IN ROUTER 2

Pool: HR	Network: 29.0.0.0/8	Excluded Address: 29.0.0.1 – 29.0.0.60
Pool: SALES	Network: 1.0.0.0/8	Excluded Address: 1.0.0.1 – 1.0.0.60
Pool: MANAGEMENT	Network: 129.23.0.0/16	Excluded Address: 129.23.0.1 – 129.23.0.60
Pool: PRODUCTION	Network: 212.25.0.0/24	Excluded Address: 212.25.0.1 – 212.25.0.60



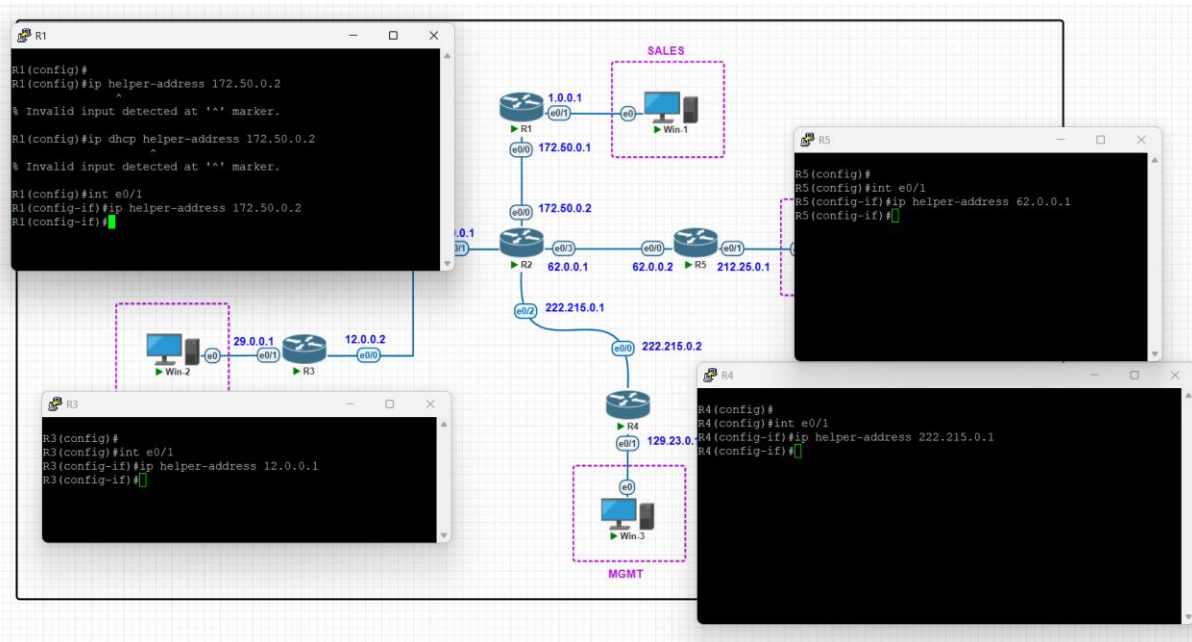
TASK 3: ADDING HELPER ADDRESS IN THE ROUTERS

R1: 172.50.0.2

R3: 12.50.0.1

R4: 222.215.0.1

R5: 62.0.0.1



TASK 4: DHCP IP ASSIGNMENTS TO PCs

PC 1 IP: 1.0.0.61

PC 2 IP: 29.0.0.61

PC 3 IP: 129.23.0.61

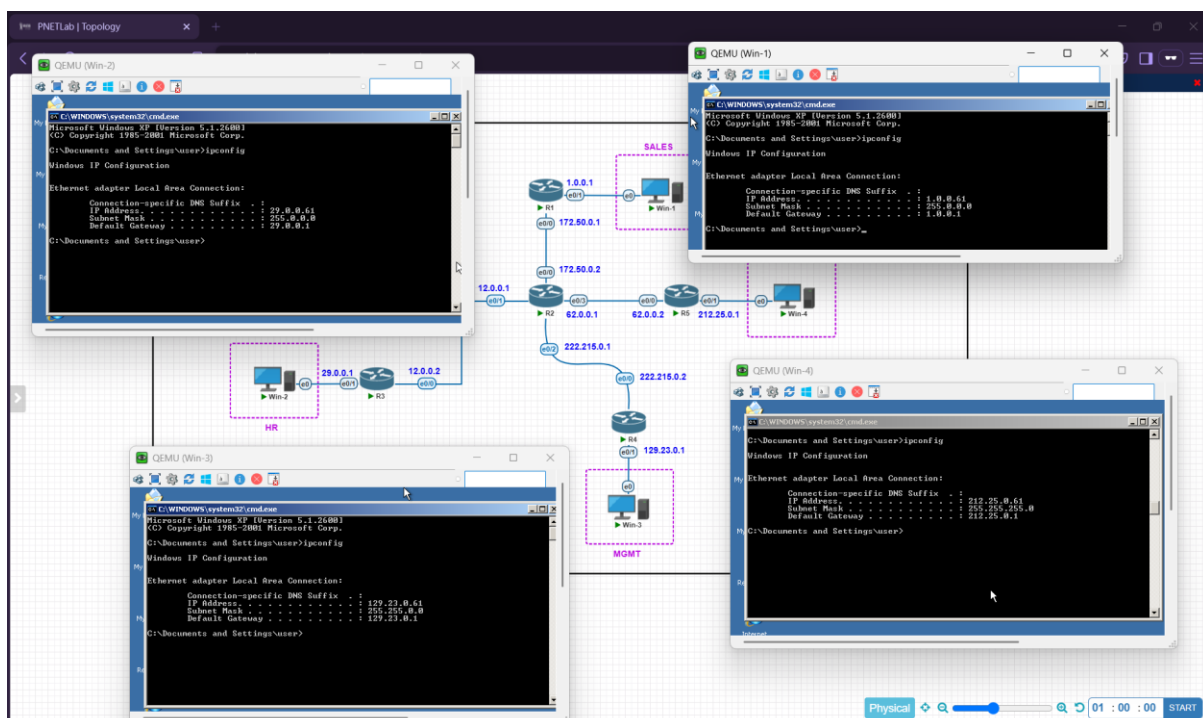
PC 4 IP: 212.25.0.61

Gateway: 1.0.0.1

Network: 29.0.0.1

Network: 129.23.0.1

Network: 212.25.0.1

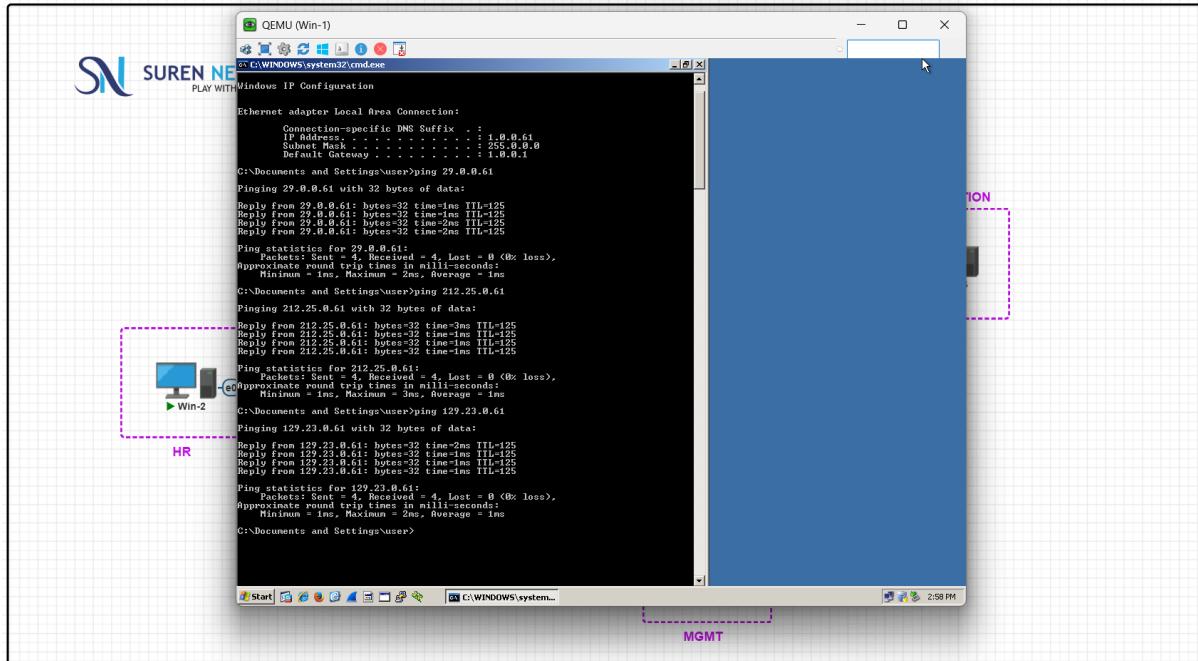


TASK 5: VERIFYING COMMUNICATION ESTABLISHMENT BETWEEN PCs

ping from 1.0.0.61 (PC1) to 29.0.0.61 (PC2) &

ping from 1.0.0.61 (PC1) to 129.23.0.61 (PC3) &

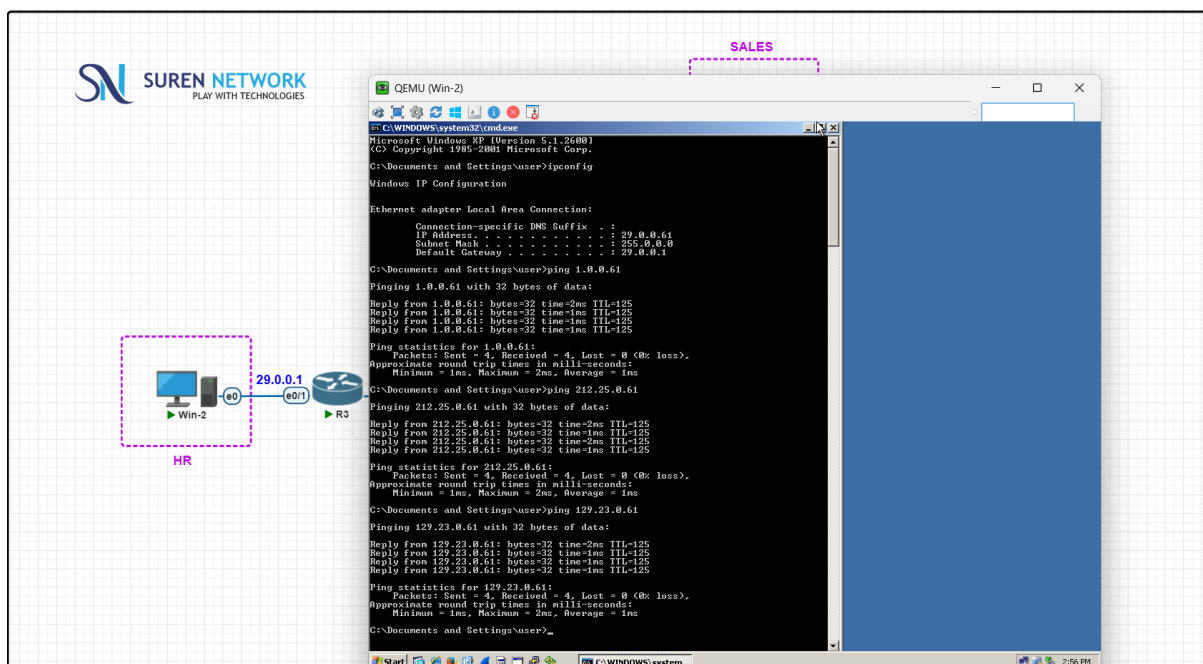
ping from 1.0.0.61 (PC1) to 212.25.0.61 (PC4) &



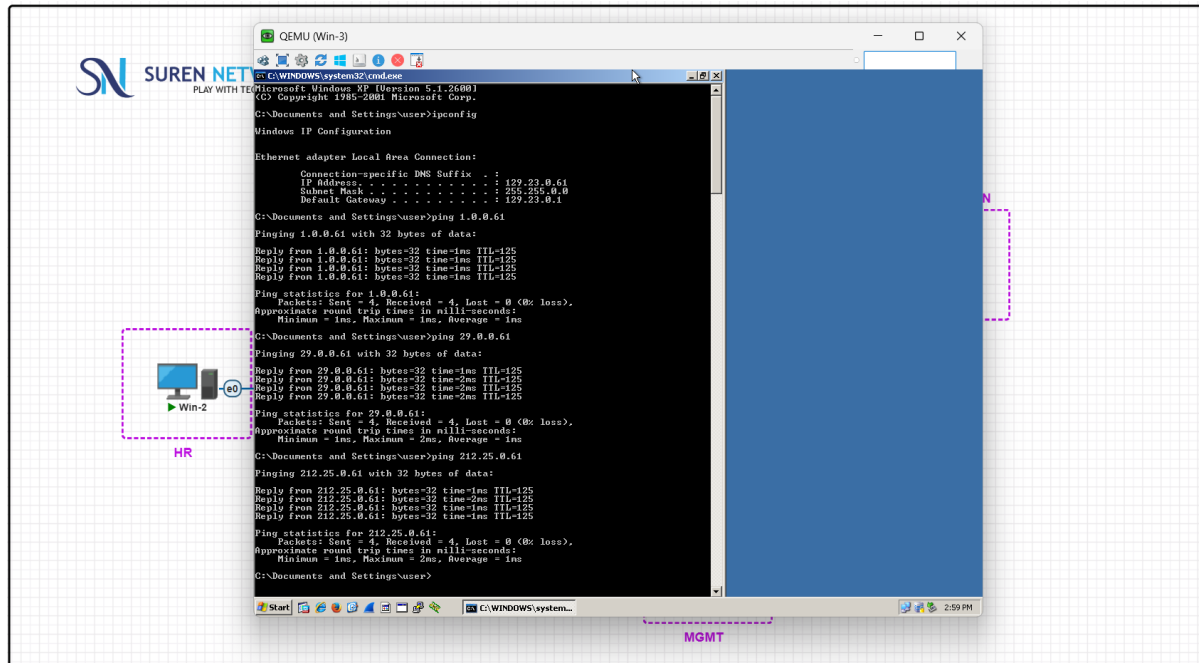
ping from 29.0.0.61 (PC2) to 1.0.0.61 (PC1) &

ping from 29.0.0.61 (PC2) to 129.23.0.61 (PC3) &

ping from 29.0.0.61 (PC2) to 212.25.0.61 (PC4) &



ping from 129.23.0.61 (PC3) to 1.0.0.61 (PC2) &
ping from 129.23.0.61 (PC3) to 29.0.0.61 (PC3) &
ping from 129.23.0.61 (PC3) to 212.25.0.61 (PC4) &



ping from 212.25.0.61 (PC4) to 1.0.0.61 (PC1) &
ping from 212.25.0.61 (PC4) to 29.0.0.61 (PC2) &
ping from 212.25.0.61 (PC4) to 129.23.0.61 (PC3) &

